Student Attendance Analytics System

Project Plan: Problem, Solution, Roles and Database Design

Core idea: Problem → Data → Analytics → Insight → Decision → Solution → Benefit

> **In one sentence:** This is not just a system that records who is present. It is a system that turns attendance data into useful knowledge, so a school can spot problems early, decide who needs help first, and check whether its actions worked.

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1. The big idea: Analytics, not just recording

Most attendance systems only save data and show a list: who was present and who was absent. In this project, attendance is only the **source of data**. The real work is the **analytics** done on that data. The system should answer these questions:

- What is happening?
- Why might it be happening?
- Which students have attendance problems?
- Is the problem serious or minor?
- Which class, subject, teacher, time or day looks unusual?
- Who should the school help first?

> That is what makes this an **Analytics System** and not an *Attendance Recording System*.

2. The problems the system solves

Problem 1: Schools know a student is absent, but not what the absence means

Example: Dara was Present 18 times, Absent 5 times and Late 3 times in a month. A teacher says "Dara is absent a lot." That is not deep enough. A good system would find a **pattern**:

- Dara is rarely absent on Monday and Friday.
- Dara is often absent on Wednesday, mostly in the first period.
- Most of those absences happen in Math.
- Over the last 4 weeks, absences have been increasing.

> **Analytics result:** "Dara has a recurring absence pattern during Wednesday morning Math classes, with attendance declining over the last four weeks."

Problem 2: The attendance rate alone can hide the truth

The usual formula is: **Attendance Rate = Present / Total Classes x 100%**. But two students can both have 90% and be very different:

Week	Student A	Student B
Week 1	100%	90%
Week 2	100%	90%
Week 3	100%	90%
Week 4	80%	90%
Week 5	70%	90%
Average	**90%**	**90%**

Student A needs more attention because the trend is going down. So the system should compare current attendance with the previous week, previous month, previous term and historical pattern. Example display:

87% attendance - down 8% compared with last month - declining trend detected.

Problem 3: Schools find problems too late

A student's attendance may go 95%, 90%, 82%, then 68% over four months. The school only reacts in month 4, although the problem started in month 2. That is **reactive** management. The system should be **proactive**:

Data → Detect pattern → Identify risk → Alert → Intervention

Warning signs the system can detect:

- Attendance falling for 3 weeks in a row
- Frequent late arrivals
- Repeated absence in the same subject
- Absences concentrated on certain days
- Growing number of consecutive absences
- A sudden change from previous behavior

From these signs the system can calculate an **Attendance Risk Score**:

Student	Attendance	Trend	Late	Pattern	Risk
Dara	91%	Down	4	Increasing	Medium
Sokha	84%	Down a lot	7	Increasing	High
Lina	96%	Stable	1	Normal	Low
Vuthy	88%	Down	2	Irregular	Medium

Problem 4: Is the problem about one student, or about the system?

If Class 10A has low attendance, a school might think the students lack discipline. But analytics may show:

- Math 95%, Khmer 94%, English 92%, **Physics 76%**

Now the question changes from "Why are 10A students absent?" to "Why is attendance low in Physics?". Digging further (teacher, period, day, room) might show that **Physics, Period 1, Monday** is the issue. The cause could be the schedule, transport, classroom location, subject difficulty or a timetable conflict.

Important: analytics should never accuse a teacher. It should only reveal patterns the school should investigate.

Problem 5: Schools don't know when and where problems happen

Time-based analytics finds hotspots. Example: by day, Wednesday is lowest (86%). Drilling into Wednesday: Period 1 is 78%, Period 2 is 91%, Period 3 is 93%. So **Wednesday Period 1 is an attendance hotspot**, something that is hard to see in a plain Excel table.

Problem 6: Not all absences mean the same thing

Two students may both have 8 absences. Student A: 6 sick, 2 family, 0 unexcused. Student B: 0 sick, 0 family, 8 unexcused. These are very different situations. The system should show the **Absence Reason Distribution** (excused, unexcused, sick, family reason, school activity, late, left early).

Problem 7: Consecutive absences matter more than totals

Student A is absent 1 day, present 1 day, absent 1 day, present 1 day. Student B is absent 4 days in a row. Totals can be equal, but Student B is more worrying. The system should send a **Consecutive Absence Alert**, for example: "Student has been absent for 3 consecutive school days", to the class teacher, attendance officer and school admin.

Problem 8: Who should be helped first?

If a school has 1,500 students and 300 are below 90%, where should the admin start? A plain table of Present/Absent does not help. The system should create an **Intervention Priority** list (Critical, High, Medium, Low) based on attendance level, declining trend, consecutive absences, unexcused absences, lateness, frequency and pattern changes.

Problem 9: Did the school's action actually work?

Suppose the school contacts a student's parents. Two weeks later the system compares attendance:

- Before 68%, after 87%: the intervention had a positive result.
- Before 78%, after 61%: attendance kept declining, so the school may need a different approach.

This creates a feedback loop: **Detect** → **Intervene** → **Measure** → **Evaluate** → **Improve**. Ordinary attendance systems rarely do this.

3. The five layers of analytics

The whole system is organized into five layers, from simple to advanced:

Layer	Question it answers	Examples
1. Descriptive	What happened?	Attendance rate, absent count, late count, attendance by class, grade and month
2. Diagnostic	Why might it be happening?	Which subject has the most absences? Which day is lowest? Which period has the most lateness? Is it getting worse?
3. Predictive / Early Warning	What may happen if the pattern continues?	Attendance 90% → 87% → 82% → 75% is flagged: High Risk, declining consistently
4. Prescriptive	What should the school consider doing?	After 3 absences in a row, suggest: review the reason, teacher follow-up, contact guardian if policy requires, monitor for 2 weeks
5. Intervention Effectiveness	Did our action help?	Before 70%, after 88%: improved by 18 percentage points

> **Two cautions.** (1) In Layer 3, use the terms **At-Risk** and **Early Warning**. Do not claim to predict that a student will drop out unless you have a proper historical dataset and model. (2) In Layer 4, the system should **support decisions, not make them**. People decide.

4. What the system analyzes and shows

Analytics groups

Group	What it includes
Student	Attendance rate, trend, absence frequency, consecutive absence, absence reason, subject attendance, day/period pattern, risk level
Class	Class attendance rate and trend, absence distribution, number of high-risk students, comparison between classes
Grade	Grade 7 vs 8 vs 9, attendance trend, risk distribution
Subject	Attendance by subject, subject-specific and period-specific patterns
Time	Day of week, week, month, term, period
Intervention	Number and type of interventions, before vs. after attendance, improvement rate, cases needing follow-up

Don't build a dashboard full of charts with no purpose

A bar chart, a line chart and a pie chart are only visuals. Real analytics is **Data** → **Analysis** → **Insight** → **Decision**. Compare how a single number becomes more useful:

Level	What the dashboard says
Weak	Attendance = 84%
Better	Attendance = 84%, down 7% from the previous month

Level	What the dashboard says
Even better	Attendance = 84%, down 7% from the previous month, with 62% of absences in first-period classes
Best	Attendance is declining, especially in first-period classes. 12 students have entered the high-risk group and should be reviewed.

Principal dashboard (homepage example)

Indicator	Example value
Overall attendance this month	91.4% (down 2.8%)
High-risk students	37
Students with consecutive absences	12
Frequently late students	46
Classes with declining attendance	5

Below this, the dashboard shows an attendance trend chart over the months and attendance by grade (for example Grade 7: 94%, Grade 10: 86%), and highlights insights such as "Grade 10 shows the largest decline this term."

Drill-down: School → Grade → Class → Student → Pattern

The principal clicks **Grade 10** and sees each class (10A 94%, 10B 91%, **10C 84%**, 10D 93%). Clicking **10C** shows: attendance 84%, trend down 9%, 8 high-risk students, most affected day Wednesday, most affected period 1, lowest subject Physics. Clicking a student opens the **Student 360-degree Profile**:

Field	Example (student Dara)
Attendance / trend	78%, down 14%
Late arrivals	9
Consecutive absence	3 days
Highest absence	Wednesday, Period 1, Physics
Risk level	High
Recent intervention	Parent contacted
After intervention	72% → 78%
Status	Improving, continue monitoring

5. Android app vs. website

The two should not have the same interface. Each has a different job.

	Android app	Website
Used by	Teachers	School administrators, principal, academic management
Purpose	Data collection and immediate action	Analytics, monitoring and decision support

	Android app	Website
Features	Login, select class, mark attendance, add absence reason, view student attendance, receive alerts, follow up students	School-wide dashboard, analytics, student risk, class/grade comparison, subject analysis, trends, alerts, intervention management, reports, export

Android = Data Collection + Immediate Action. Website = Analytics + Monitoring + Decision Support.

6. Problem statement, solution and benefits

Core problem statement

> Schools may collect large amounts of student attendance data, but often lack an analytical mechanism to transform that data into meaningful insights about attendance patterns, declining trends, at-risk students, and intervention effectiveness. As a result, problems are detected late and management decisions may not be based on enough evidence.

Please avoid making "paper records are slow" the main problem. It is too basic and many projects already cover it.

Proposed solution

> The Student Attendance Analytics System collects and analyzes attendance data to find attendance patterns, trends, risk indicators and early-warning signals. It provides dashboards, reports and analytical insights that help teachers and school management prioritize which students to follow up and evaluate results after interventions.

The system pipeline

Attendance data → Data collection → Data processing → Attendance analytics → Pattern detection → Trend detection → Risk identification → Early warning → School intervention → Monitor result → Measure intervention effectiveness

The system does not stop at "Save Attendance". It goes on to answer: *"What should the school understand and do with this data?"*

Benefits

Benefit	Meaning
Early detection	Find struggling students before attendance drops sharply
Better decisions	Admins rely on data instead of feelings or assumptions
Hidden patterns	Reveal day, period, subject, class and trend patterns that a plain list hides
Prioritized intervention	Show who needs attention first
Measured intervention	Show what the school did and whether it worked
School-level planning	Show where and when problems occur to improve school management

Be careful not to over-claim. Don't write "Our system will identify why students are absent" - attendance data alone cannot show the real "why". Write instead: "The system identifies attendance patterns and factors that may warrant further investigation." Also avoid "Our AI predicts which students will drop out." Use **Attendance Risk Detection / Early Warning**, which is more defensible.

7. How teachers record attendance

The recording flow should be as short and simple as possible:

Step	What happens
1	Teacher logs in
2	System shows today's teaching sessions (for example: Class 12A, 7:00 AM, Mathematics)
3	System automatically shows the student list for that class
4	Teacher marks each student Present or Absent
5	Teacher saves (submits) the attendance
6	System shows a confirmation / summary

Key principle: the teacher should never have to choose students manually. The system knows which class, subject and period the teacher teaches and loads the right student list automatically.

8. User roles and permissions

There are three roles. Permissions are defined clearly to keep attendance data accurate and secure.

What each role does

Role	Purpose	Main responsibilities
Admin	System administrator: prepares the system	Create and disable user accounts; enter students, teachers, classes and subjects; assign teachers to classes and subjects; set schedules; manage academic years; review reports and audit logs
School Director	Monitoring and decision-making	View the school-wide dashboard; see attendance by class, student and period; find students with frequent absence; compare classes (12A, 12B, 12C); see classes that have not finished recording; download reports. Cannot edit attendance directly by default
Teacher	Attendance recorder	Log in; view own schedule; open a class card (for example 12A, 7:00 AM, Math); mark Present/Absent; save; view attendance history and analytics for own classes; correct own records within the time limit

Permission table

Permission	Admin	School Director	Teacher
Manage user accounts	Full access	No	No
Manage student records	Create, edit, archive	View all	View assigned classes
Manage teachers	Full access	View all	View own profile
Manage classes and subjects	Full access	View all	View assigned classes
Manage teaching schedules	Full access	View all	View own schedule
Record attendance	Only if authorized	No by default	Assigned classes only
Correct submitted attendance	Authorized corrections	View; oversee logs	Own records within time limit
View analytics	All school	All school	Assigned classes
Export reports	All school	All school	Assigned classes, if enabled
View audit logs	Full access	Read-only, school-wide	Own activity, if enabled

Attendance correction rules (proposed starting values, adjustable later)

- **Correction window: 24 hours.** A teacher can correct attendance they saved within 24 hours.
- **Audit log for every correction:** who changed it, when, old status, new status and reason.
- **After 24 hours:** the teacher cannot edit directly. They must submit a correction request to someone with authority, and the old history is never deleted.
- The backend (not just the app screen) must check this time limit so nobody can bypass it.

Security rules the system must follow

Rule	Meaning
Authentication	Users must log in before accessing any data
Role-Based Access Control	Basic permissions depend on Admin, School Director or Teacher
Assignment-based access	Teachers can only access classes and subjects assigned to them
Attendance validation	Only Present or Absent; no accidental duplicate record for the same student and session
Correction history	Every correction stores the editor, time and old/new values
Server-side permission checks	The backend checks permission on every read or edit; hiding a button in the app is not enough

9. Database design (PostgreSQL)

The system uses **PostgreSQL** with one shared backend and one shared database for both the Android app and the website. The key design idea is to separate two things:

- **Attendance Session** = one lesson on a specific date and time (for example 12A, Math, 7:00 AM on a given day).
- **Attendance Record** = one student's Present/Absent status inside that session.

This separation lets the system calculate attendance rates and analyze history correctly.

Table groups

Group	Tables	Purpose
Access control	roles, users, teachers	Login and permissions
School structure	academic_years, classes, subjects, students	Core school data
Assignments	student_enrollments, teaching_assignments, schedules	Who studies and teaches what, and when
Attendance	attendance_sessions, attendance_records	Recording and tracking
Accountability	audit_logs, correction_requests	History and post-deadline corrections

Main tables

Table	What it stores	Key fields
roles	The three roles	role_id, role_name
users	Login accounts	user_id, role_id, username, password_hash, is_active
teachers	Teacher info linked to a user	teacher_id, user_id, teacher_code, full_name
academic_years	School years	year_name, start_date, end_date
students	Student information	student_code, full_name, gender, is_active
classes	Classes such as 12A, with shift	class_name, shift, academic_year_id
subjects	Math, Physics, Khmer...	subject_code, subject_name

Table	What it stores	Key fields
student_enrollments	Which student is in which class in which year (keeps history when students move up)	student_id, class_id, enrolled_from, enrolled_to
teaching_assignments	Links teacher + class + subject	teacher_id, class_id, subject_id, effective dates
schedules	Day and time of each lesson	assignment_id, day_of_week, start_time, end_time
attendance_sessions	One lesson on one date	schedule_id, session_date, recorded_by, submitted_at, status (DRAFT or SUBMITTED)
attendance_records	One student's status per session	session_id, student_id, attendance_status, marked_by
correction_requests	Requests to change attendance after 24 hours	record_id, old/requested status, reason, reviewer
audit_logs	History of important changes	user_id, action, old_values, new_values, reason, created_at

Database rules that must be enforced

Rule	Requirement
Unique attendance	No two records with the same session and student
Valid status	Only Present and Absent allowed
Teacher authorization	The teacher must be assigned to that class and subject
Correction deadline	The backend checks the 24-hour window
Audit integrity	Correction history cannot be edited or deleted in normal use
Enrollment validation	The student must be enrolled in the class of that session
Unsubmitted sessions	Draft sessions are never counted as absences

Correction safeguards (triggers)

The plan includes two PostgreSQL triggers as an extra safety layer:

- **Guard trigger:** blocks attendance changes unless the user is authenticated and active, the session is already submitted, and (for teachers) it is their own session within 24 hours. Admin and School Director corrections are allowed but are always logged.
- **Audit trigger:** automatically writes an audit log entry whenever an attendance status changes.

These triggers are a starting safeguard, not the full security layer. The backend must set the logged-in user ID itself after authentication (never trust an ID sent from the browser or app), and direct table writes should be limited to the backend service account.

10. Analytics calculation and development plan

How analytics uses the database

Data flows in this order: **Teaching Schedule** (class, teacher, subject, time) → **Attendance Session** (one class meeting on a date) → **Attendance Records** (one Present/Absent per enrolled student) → **Analytics**

and Reports (rates, patterns, class comparisons).

Attendance rate formula

> **Attendance Rate = Present Records / Eligible Submitted Attendance Records x 100%**
> Define the analysis period and the number of sessions a student should attend. Do not count sessions that have not been submitted.

The starter database view *student_attendance_analytics* counts present, absent and attendance rate per student, class and academic year, using submitted sessions only. For monthly trends, class comparisons and teacher-level reports, additional queries or views are needed.

Important limitation: the view counts only the records that exist. If a student has no record for a session, the percentage could be inflated. So when a teacher submits a session, the system must verify the roster and create a Present or Absent record for **every** eligible student before marking the session as submitted.

Implementation checklist

#	Task
1	Create the PostgreSQL database and manage changes through migrations
2	Seed the three roles: ADMIN, SCHOOL_DIRECTOR, TEACHER
3	Use one shared backend and database for Android and website
4	Keep student enrollment and class history across academic years
5	Validate schedule, teacher assignment, class, subject and enrollment consistency
6	Enforce one attendance record per student per session
7	Allow only Present and Absent
8	Enforce the 24-hour teacher correction rule on the server and database
9	Record corrections and privileged changes in audit logs
10	Build a post-deadline correction request and approval workflow
11	Build analytics that exclude draft sessions and handle missing records
12	Test role permissions, concurrent submissions, correction deadlines and audit integrity

Recommended development order

Step	Work
1	Finalize and test the database migrations
2	Implement authentication and role-based authorization in the shared backend
3	Implement student enrollment, teaching assignments and schedules
4	Implement attendance session creation, submission and correction
5	Build analytics queries and reports
6	Connect the Android app and the website to the same API
7	Test security and attendance calculations with realistic school data

> **Recommendation:** finish the database schema and its integrity rules before designing all the app screens. This reduces rework and keeps attendance data reliable as the school adds classes, teachers, subjects and academic years.